Introducing a New Method for Simulating Kinetic Models

- Faster & More Accurate than NEURON's current solver
- For models which are Linear & Time-Invariant
  - Markov Models of Ion Channels
- Proof of concept implemented for NMODL files

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What is a Linear & Time-Invariant Differential Equation?

Problem: $dx/dt = A * x$

- Where X is a variable
- Where A is a constant

Solution: $x_{t+dt} = x_t * \exp(A * dt)$

But what if X is a vector and A is a matrix?

The exponential function is defined for matrices.

How to compute the matrix?

- Compute ahead of time

  “exp(matrix)” is expensive to compute!

- The matrix is a function of the input

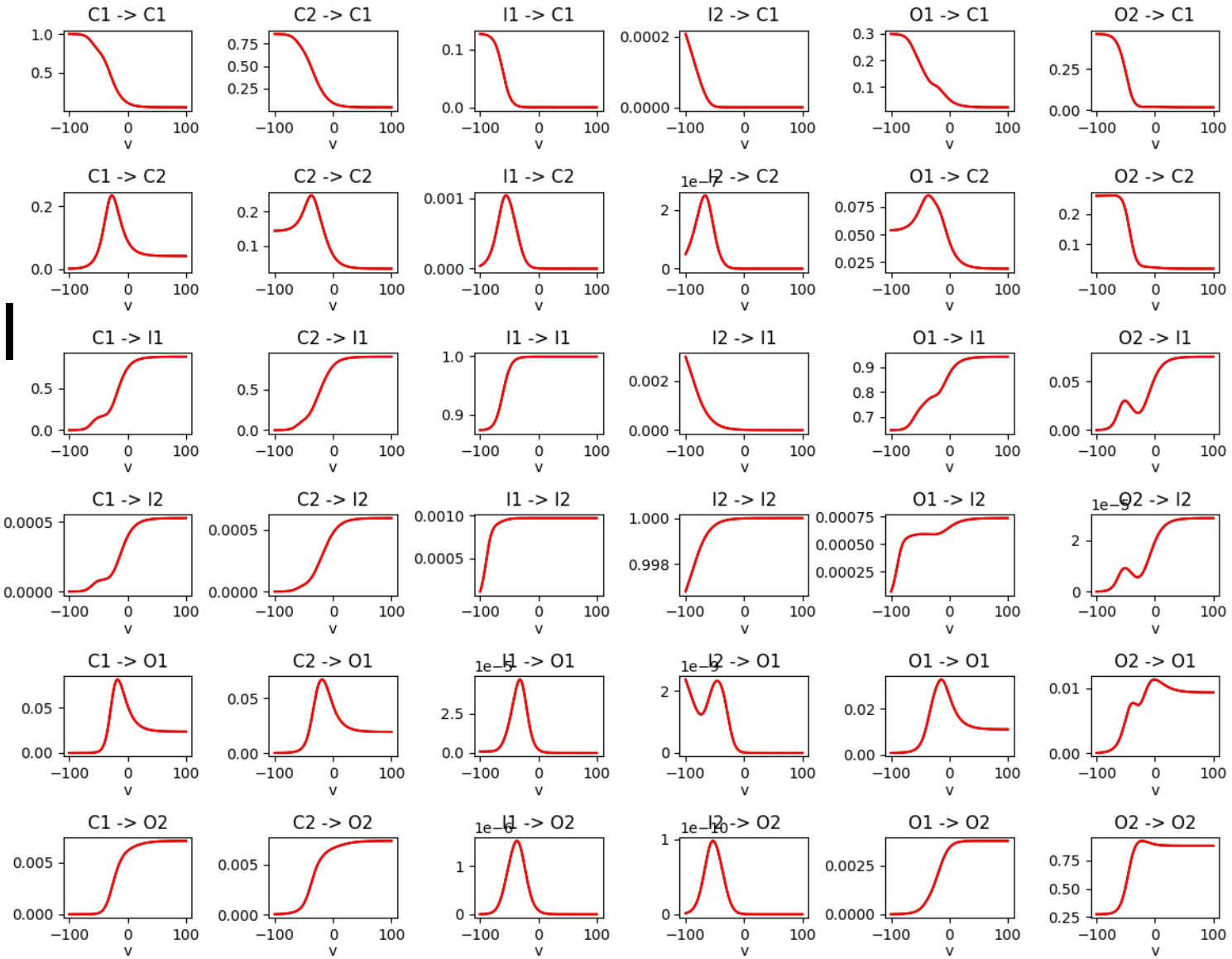
  “exp(A(input) * dt)”

  Assume the input is constant for the duration of each time step

Let's look at an example …

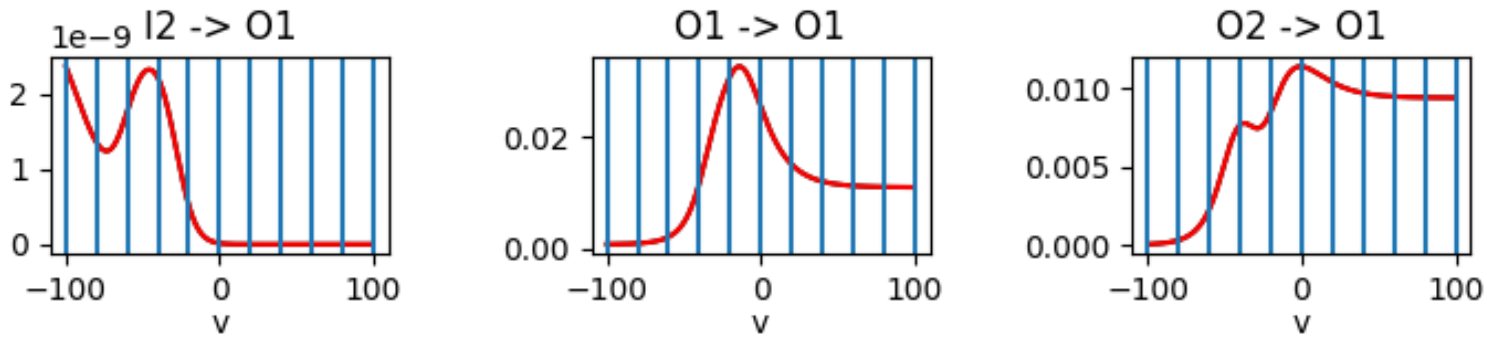
Example: Kinetic Model of Nav1.1 with 6 states

These plots shows the fraction of each state that transitions to each other state, as a function of voltage, over the course of one time step.



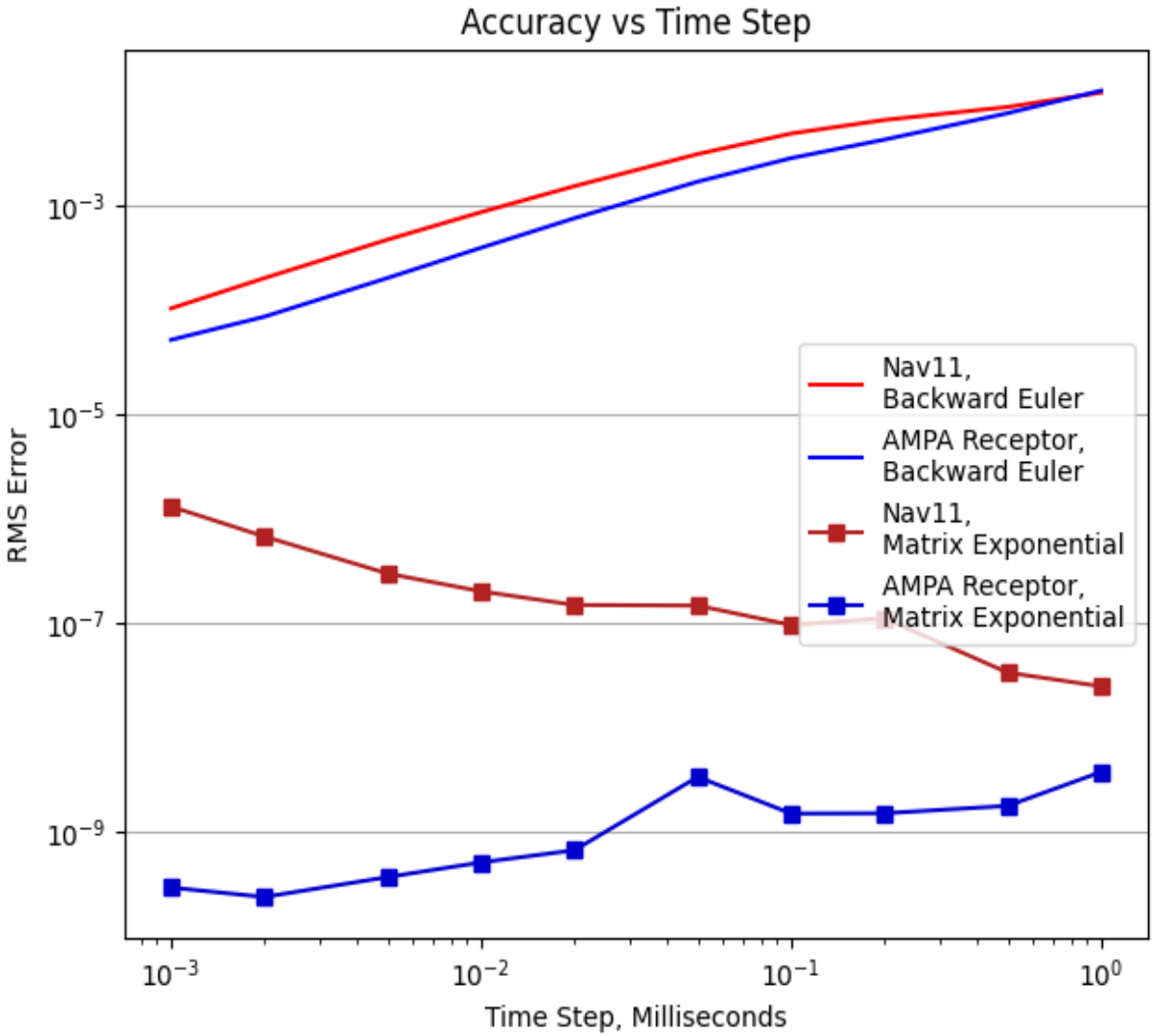
How to approximate the matrix?

- Divide the input space into small pieces
- Fit a polynomial approximation to each piece



Accurate

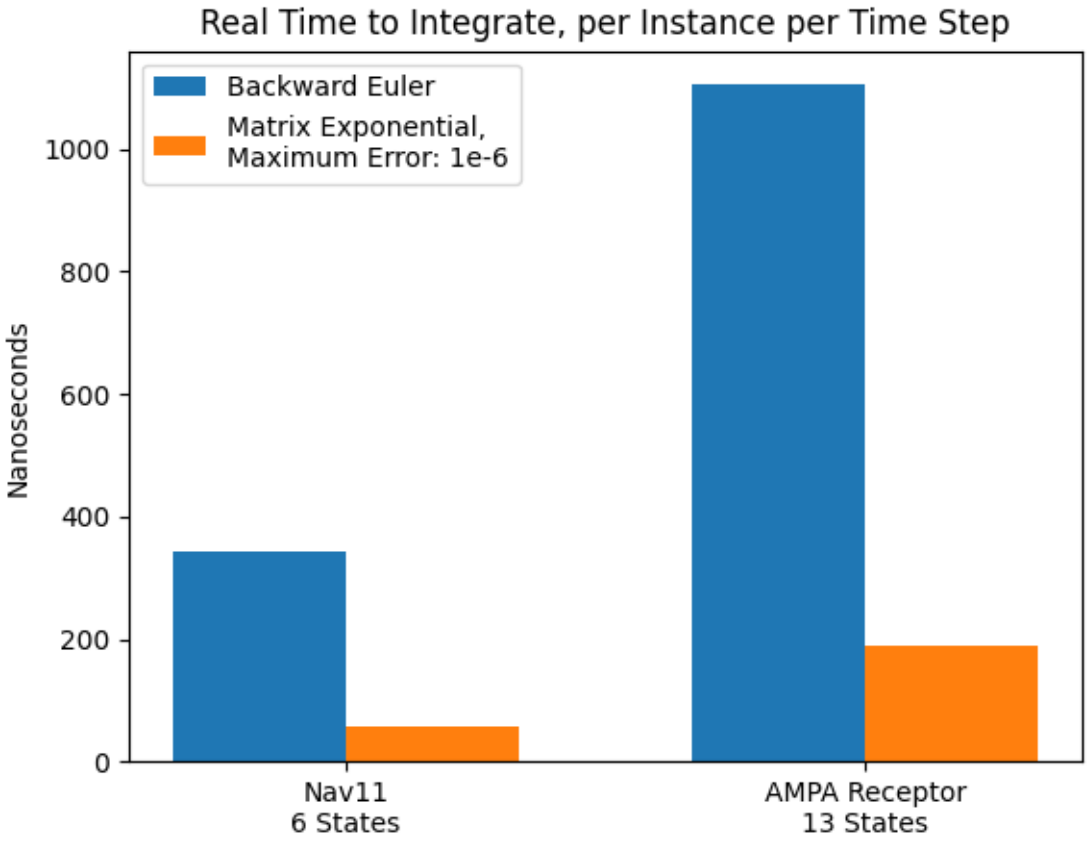
- The matrix exponential is an exact solution
- Its approxiation is the only source of error
- Automatically measure and control the error by:
  - Increasing the number of input bins
  - Increasing the order of the polynomials



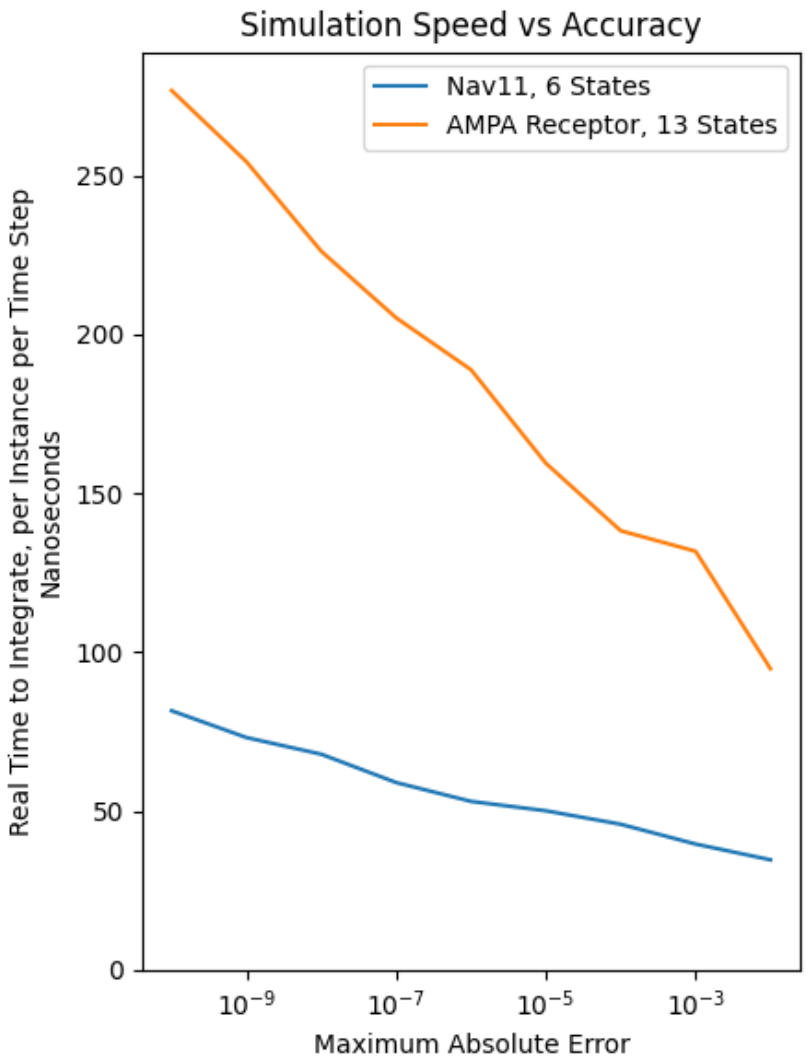
Fast

Note: the performance of the matix exponential method relies on computing many instances in large batches.

This is because loading the approximation from RAM into CPU cache is costly.



Speed vs Accuracy



Pros & Cons

Pros

- Fast
- Accurate

Cons

- Only 1 or maybe 2 inputs allowed
- Variables become constant
  - Temperature
  - Time Step
  - Parameters
  - ASSIGNED block
- Complicated to implement

- Reference:

  Exact digital simulation of time-invariant linear systems with applications to neuronal modeling. Rotter S, Diesmann M (1999). https://doi.org/10.1007/s004220050570

- Source code:

  https://github.com/ctrl-z-9000-times/lti_sim